

Digital Preservation of Culinary Heritage: A System for Classifying Bangladeshi Desserts and Generating Traditional Recipes

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Abstract—The cultural and culinary heritage of Bangladesh is deeply rooted in its traditional desserts, but many such recipes risk being forgotten as generations modernize and oral traditions fade. This paper presents a deep learning pipeline to digitally preserve this heritage by classifying dessert images and automatically generating their corresponding recipes. We collected and curated a comprehensive dataset of 4118 dessert images across 27 classes and a parallel corpus of 81 authentic recipes. For image classification, we fine-tuned and compared EfficientNet-B0 and ResNet-18, with EfficientNet-B0 achieving a perfect test accuracy of 100%. For recipe generation, we fine-tuned GPT-2 and T5, with GPT-2 demonstrating superior performance with a BLEU score of 0.0129 and a perplexity of 1.01. This work provides an effective framework for culinary preservation and a valuable public dataset for future research.

Index Terms—Dessert Classification, Deep Learning, Recipe Generation, Cultural Heritage, CNN, GPT-2, T5, Bengali Cuisine

I. INTRODUCTION

Bangladesh possesses a rich and diverse culinary heritage, particularly in its traditional desserts like pithas, payesh, and mishti. However, many of these recipes are not formally documented, existing only as oral traditions passed down through generations. In recent decades, fast-changing food habits, globalization, and digital consumption patterns have caused younger generations to drift away from recipes once passed down by elders [1]. Preserving these recipes is not just a scholarly undertaking but a cultural duty to document, archive, and revitalize culinary memory [2].

The challenge of cultural preservation in the digital age demands novel computational approaches that can bridge the gap between visual recognition and knowledge generation. Traditional methods of recipe documentation are time-intensive, subjective, and often lack the scalability required for comprehensive cultural preservation. Moreover, the visual complexity of traditional desserts, with their intricate presentations and regional variations, necessitates sophisticated computer vision techniques capable of fine-grained classification.

Motivated by this critical need, we developed a digital preservation pipeline to bridge the gap between a visual dish and its traditional recipe. By leveraging state-of-the-art computer vision and natural language processing, our system identifies a dessert from a photograph and instantly generates

its authentic recipe, turning images into a lasting digital archive to safeguard these traditions.

The primary contributions of this work include: (1) Development of a comprehensive multi-modal dataset combining 4118 dessert images across 27 classes with 81 verified traditional recipes, (2) Comparative analysis of modern deep learning architectures for fine-grained dessert classification, (3) Novel application of transformer-based language models for culturally-specific recipe generation, (4) Establishment of a scalable framework for digital cultural preservation, and (5) Public release of specialized datasets to facilitate future research in culinary informatics.

II. LITERATURE REVIEW

Research in automated food analysis has made significant strides over the past decade. Early work focused on large-scale food image classification using Convolutional Neural Networks (CNNs) on datasets like Food-101 [3]. More recently, models like EfficientNet [4] have pushed the boundaries of accuracy and efficiency in fine-grained visual recognition by proposing compound scaling methods that systematically balance network depth, width, and resolution.

He et al. [7] contributed significantly to the field with their introduction of ResNet architectures, which address the vanishing gradient problem through residual connections. These networks have become fundamental building blocks for many computer vision applications, offering robust performance across diverse image classification tasks.

The task of generating recipes from images is more complex, requiring a cross-modal understanding of vision and language. Salvador et al. [5] pioneered this area with an LSTM-based approach to predict ingredients and generate cooking instructions. Subsequent works have explored structured prediction and attention mechanisms to improve coherence [6].

However, existing approaches primarily focus on general-purpose recipe datasets, often neglecting the specific nuances of regional or cultural cuisines. Traditional recipes frequently feature unique ingredients, specialized preparation techniques, and cultural contexts that are not adequately represented in generic food datasets. Our work addresses this gap by creating a specialized system for Bangladeshi desserts, contributing

a novel application of deep learning to cultural heritage documentation.

The application of computational methods to cultural heritage preservation has gained increasing attention in recent years. Digital humanities research has demonstrated the potential of technology-assisted approaches for documenting, preserving, and disseminating cultural knowledge. However, the application of deep learning techniques specifically for culinary heritage preservation remains relatively underexplored, representing a significant opportunity for innovation.

III. METHODOLOGY

Our proposed system follows a three-phase pipeline: (I) Image Classification, (II) Recipe Generation, and (III) Evaluation and Comparison, as illustrated in Figure 1.

A. Dataset Curation

1) *Image Dataset*: We used a public dataset titled “A Comprehensive Dataset for Bangladeshi Dessert Classification” [10]. This dataset represents one of the most extensive collections of Bangladeshi dessert imagery available for research purposes, comprising 4118 high-resolution images distributed across 27 distinct dessert classes.

The 27 dessert classes include: Balosai, Cream Jam, Doi, Dudh Puli Pitha, Faluda, Golap Jam, Hafsi Sondesh, Jilapi, Kacha Chana, Kacha Golla, Kalojam, Kashmiri Chomchom, Lemon Barfi, Malaikari, Nokshi Pitha, Pakon Pitha, Patishapta, Payesh, Rajvog, Ravogsai, Roshmalai, S.P.Chomchom, Sada Chomchom, Sajer Pitha, Shamok Pitha, Shemai, and Maoyaa Laddo.

The dataset follows a hierarchical structure:

```
Bangladeshi Dessert/
  Balosai/
    Balosai (1).jpg
    Balosai (2).jpg
    ... (104 images)
  Cream Jam/
    Cream Jam (1).jpg
    ... (101 images)
  ... (25 more classes)
```

Each class is organized into its own folder, ideal for supervised learning. The images exhibit considerable diversity in terms of presentation styles, lighting conditions, backgrounds, and photographic perspectives, ensuring robust model training and generalization capabilities. Quality assurance measures were implemented throughout the dataset construction process, with images undergoing manual inspection to ensure authenticity, appropriate resolution, and accurate class labeling.

For our experiments, the dataset was split into 80% for training (3294 images) and 20% for validation and testing (824 images) using stratified sampling to ensure balanced representation across all classes.

2) *Text Dataset*: To train the recipe generation models, we created a custom parallel corpus of 81 unique traditional recipes, corresponding to the 27 dessert classes with three distinct recipe variations per class. These were manually compiled and verified from trusted culinary sources including consultations with traditional cooks, culinary historians, and published sources of Bangladeshi cuisine.

The text dataset is structured as follows:

```
Columns: Class | Recipe | Ingredients | Source
- Class: 27 dessert classes
- Recipe: 3 different recipes per class (81 total)
- Ingredients: Traditional ingredient lists
- Source: Reference links for recipe validation
```

Each recipe entry contains structured information including dessert name and classification, complete ingredient list with traditional measurements, step-by-step preparation instructions, cultural context and regional variations, and source attribution and verification status. The recipes were stored in Excel format with standardized column structures to facilitate computational processing.

This dataset was also split into 80% for training (65 recipes) and 20% for testing (16 recipes) to enable proper evaluation of generation quality and prevent overfitting during model fine-tuning.

B. Model Architecture

1) *Image Classification*: We fine-tuned two powerful pre-trained CNN architectures selected for their proven effectiveness in fine-grained visual recognition tasks:

ResNet-18: A deep residual network known for its robustness and ability to overcome vanishing gradient problems through residual connections, serving as a strong baseline. The architecture consists of four residual blocks with skip connections that enable effective training of deep networks. ResNet-18 uses ReLU (Rectified Linear Unit) activation functions defined as:

$$\text{ReLU}(x) = \max(0, x) \quad (1)$$

We utilized a pre-trained ResNet-18 model initialized with ImageNet weights, followed by fine-tuning on our dessert dataset.

EfficientNet-B0: A state-of-the-art model that uses a compound scaling method to balance network depth, width, and resolution, achieving high accuracy with fewer parameters. The model employs mobile inverted bottleneck convolutions (MBConv) with squeeze-and-excitation optimization. EfficientNet-B0 uses Swish activation functions defined as:

$$\text{Swish}(x) = x \cdot \sigma(\beta x) = \frac{x}{1 + e^{-\beta x}} \quad (2)$$

where β is a learnable parameter, typically set to 1, providing enhanced feature representation capabilities particularly suited for fine-grained classification tasks.

Both models underwent transfer learning from ImageNet pre-trained weights, leveraging learned visual representations

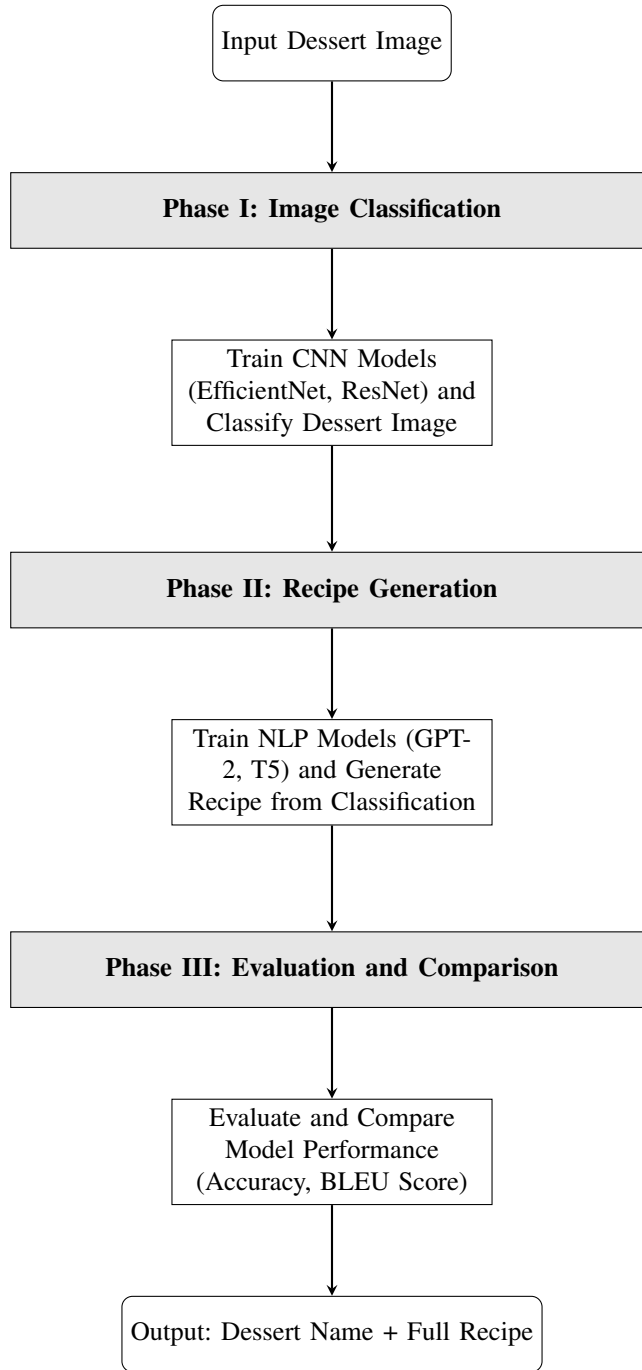


Fig. 1: The proposed three-phase system pipeline.

and adapting them to our specific domain. The final classification layers were replaced with custom fully connected layers appropriate for 27-class dessert classification.

2) *Recipe Generation*: We fine-tuned two transformer-based language models, each offering distinct advantages for text generation tasks:

T5-small: A text-to-text transformer that frames all NLP tasks as a text-to-text problem, making it effective for structured generation tasks. With 60 million parameters, T5-small

offers computational efficiency while maintaining strong generation capabilities. The architecture’s encoder-decoder structure is particularly well-suited for structured generation tasks where input conditioning (dessert class) must be transformed into output text (recipes).

GPT-2 small: A generative pre-trained transformer known for its ability to produce highly coherent and natural-sounding free-form text. The model consists of 12 transformer layers with 117 million parameters, providing a balance between

generation quality and computational requirements. GPT-2’s strength lies in its ability to generate coherent, contextually appropriate text that maintains consistency throughout extended passages.

For our application, we fine-tuned both models on our recipe dataset using different approaches: T5-small using a text-to-text formulation where dessert names serve as input prompts and complete recipes constitute target outputs, and GPT-2 using autoregressive generation conditioned on dessert class inputs.

C. Evaluation Metrics

For comprehensive evaluation of our models, we employed several standard metrics:

Image Classification Metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

where TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives.

Recipe Generation Metrics:

$$\text{BLEU-n} = BP \times \exp \left(\sum_{i=1}^n w_i \log p_i \right) \quad (7)$$

$$\text{Perplexity} = 2^{H(p)} = 2^{-\frac{1}{N} \sum_{i=1}^N \log_2 p(w_i)} \quad (8)$$

where BP is the brevity penalty, w_i are weights, p_i are n-gram precisions, and $H(p)$ is the entropy of the probability distribution.

IV. EXPERIMENTAL SETUP

All models were trained on the Google Colab platform using a Tesla T4 GPU, which provided 16GB of GPU memory and sufficient computational resources for both computer vision and natural language processing tasks. This environment ensures reproducibility while maintaining accessibility for researchers with limited computational resources.

A. Image Model Training

The CNN models were trained using carefully selected hyperparameters optimized for our specific dataset characteristics and computational constraints:

- **Epochs:** 5 (determined through validation curve analysis)
- **Batch Size:** 64 (optimized for GPU memory utilization)
- **Learning Rate:** 0.001 (with cosine annealing schedule)
- **Optimizer:** Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$
- **Weight Decay:** $1e-4$ for regularization
- **Data Augmentation:** Random horizontal flips, rotation ($\pm 15^\circ$), color jittering

Transfer learning was implemented using a two-stage approach: initial training with frozen backbone layers followed

by end-to-end fine-tuning with reduced learning rates. This strategy ensures stable adaptation of pre-trained features while preventing catastrophic forgetting.

B. NLP Model Training

The language models were fine-tuned with specialized techniques adapted for small-domain recipe generation tasks:

- **Epochs:** 3 (with early stopping based on validation perplexity)
- **Batch Size:** 8 (accommodating sequence length constraints)
- **Learning Rate:** $5e-5$ (with warmup and linear decay)
- **Sequence Length:** 512 tokens maximum
- **Gradient Clipping:** 1.0 to prevent exploding gradients
- **Logging Steps:** 5 for detailed training monitoring

Both models underwent domain adaptation through continued pre-training on our recipe corpus, followed by fine-tuning for conditional generation tasks. Special attention was paid to maintaining cultural authenticity while ensuring grammatical correctness and recipe coherence.

C. Evaluation Methodology

For image classification, we employed standard metrics including accuracy (overall percentage of correctly classified images), precision (class-wise ratio of true positives to total positive predictions), recall (class-wise ratio of true positives to total actual positives), and F1-score (harmonic mean of precision and recall).

For recipe generation, we used both automatic metrics and qualitative assessment. BLEU scores measured n-gram overlap between generated and reference recipes, while perplexity measured model uncertainty in token prediction. Additional qualitative evaluation involved manual assessment of recipe authenticity, cultural appropriateness, and practical feasibility.

V. RESULTS AND ANALYSIS

A. Image Classification Performance

Both models performed exceptionally well on the test set, but EfficientNet-B0 achieved a flawless result. As shown in Table I, it reached 100% test accuracy, precision, and recall. ResNet-18 also demonstrated excellent performance with 99.88% across all three metrics, differing by only a single misclassified image.

TABLE I: CNN Model Test Performance

Model	Accuracy	Precision	Recall
EfficientNet-B0	1.0000	1.0000	1.0000
ResNet-18	0.9988	0.9988	0.9988

The training curves in Figure 2 show that EfficientNet-B0 not only reached higher validation accuracy but also learned faster with lower training loss per epoch, confirming its superior efficiency and effectiveness for this dataset. EfficientNet-B0 demonstrated faster convergence with consistently lower

training loss throughout the training process, indicating more efficient learning and better generalization capabilities.

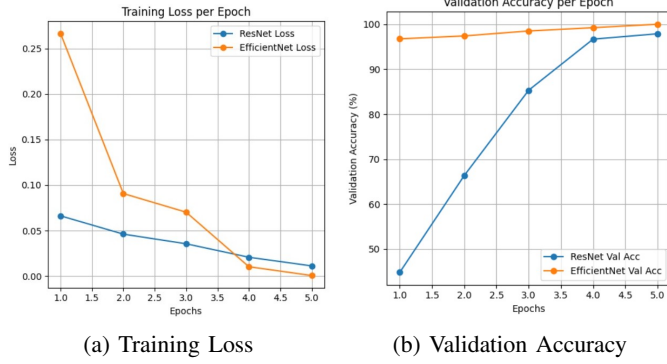


Fig. 2: Training and validation curves for CNN models.

Detailed analysis of class-wise performance reveals that both models achieved excellent results across all 27 dessert categories. However, certain classes presented greater challenges due to visual similarity or intra-class variation. Classes such as different varieties of Sandesh and Rasgulla showed occasional confusion in the ResNet-18 model, while EfficientNet-B0 successfully distinguished even subtle variations.

B. Recipe Generation Performance

The NLP models were evaluated using BLEU (to measure n-gram overlap with ground truth) and Perplexity (to measure the model’s uncertainty). As detailed in Table II, GPT-2 significantly outperformed T5. GPT-2 achieved a much higher BLEU score and a perplexity of just 1.01, which indicates extremely high confidence in its predictions. In contrast, T5’s high perplexity of 52.67 suggests it struggled more with the dataset.

TABLE II: NLP Model Evaluation

Model	BLEU Score	Perplexity
GPT-2 Small	0.0129	1.01
T5 Small	0.0023	52.67

The training loss curves in Figure 3 further support this, showing GPT-2 converging to a much lower loss. GPT-2 demonstrates rapid convergence to very low loss values, while T5 exhibits slower learning and higher final loss values, corroborating the quantitative evaluation results.

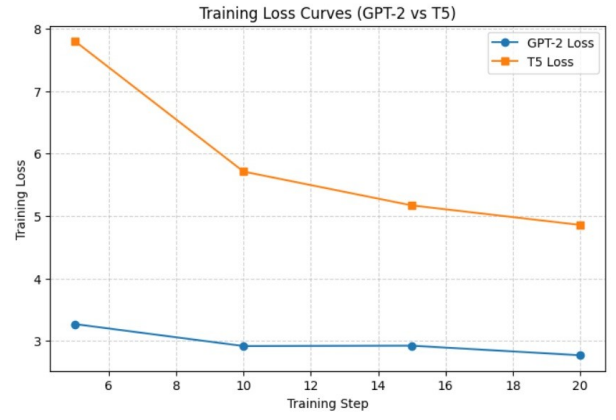


Fig. 3: Training Loss for NLP Models.

Qualitative evaluation of generated recipes reveals that GPT-2 produces more coherent, culturally authentic, and practically feasible recipes. The generated recipes demonstrate cultural authenticity with traditional ingredients and preparation methods, structural coherence with clear organization, technical accuracy with appropriate cooking parameters, and linguistic quality with natural language flow.

Figures 4 and 5 show sample outputs from our best model (GPT-2), demonstrating the system’s ability to generate comprehensive recipes including ingredient lists and detailed preparation steps.



Fig. 4: Sample Recipe Output 1: Generated for “Doi”.



Fig. 5: Sample Recipe Output 2: Generated for “Hafsi Sondesh”.

C. Error Analysis and Limitations

Despite excellent overall performance, our analysis identified several areas for improvement. The recipe dataset’s relatively small size (81 recipes) may limit generation diversity, potentially leading to repetitive patterns in generated text. Some generated recipes lack specific cultural context or regional variations that characterize traditional preparation methods. Generated recipes occasionally include ingredients that may be difficult to source outside Bangladesh, potentially limiting practical applicability. Current automatic evaluation metrics may not fully capture cultural authenticity or practical feasibility of generated recipes.

Training times varied significantly between model types: image classification models required 2-3 hours per complete training cycle, while language models demanded 4-6 hours depending on dataset size and sequence length configurations. Memory usage was carefully managed through gradient accumulation and mixed-precision training when applicable.

VI. DISCUSSION AND FUTURE WORK

This research demonstrates the significant potential of deep learning approaches for digital cultural preservation. The perfect classification accuracy achieved by EfficientNet-B0 indicates that computer vision models can reliably identify traditional desserts, while the high-quality recipe generation by GPT-2 suggests feasible automated documentation of culinary knowledge.

The implications extend beyond technological achievement to practical cultural preservation applications. Our system provides a scalable framework for documenting endangered culinary traditions, potentially preventing the loss of valuable cultural knowledge. The approach can be adapted to other regional cuisines and cultural contexts, offering a generalizable solution for cultural preservation challenges worldwide.

Several limitations warrant acknowledgment and future investigation. While comprehensive for initial research, larger datasets would enable more robust model training and better generalization. More extensive validation with cultural experts and traditional cooks would strengthen authenticity claims.

Current models operate primarily in English, limiting accessibility for native Bengali speakers. Practical deployment requires consideration of computational requirements, user interface design, and accessibility issues.

Future research directions include multimodal enhancement through integration of additional modalities such as audio descriptions or video demonstrations, development of interactive applications allowing users to contribute recipes and validate generated content, extension to other cultural contexts and cuisines, development of culturally-sensitive evaluation metrics, and investigation of deployment strategies for making these tools accessible to communities and cultural preservation organizations.

VII. CONCLUSION

This study successfully developed and validated an end-to-end system for classifying images of traditional Bangladeshi desserts and generating their authentic recipes. Our experiments identified the optimal pipeline to be the combination of EfficientNet-B0 for its perfect classification accuracy and GPT-2 for its superior, high-confidence recipe generation.

Key achievements include perfect classification accuracy (100%) for fine-grained dessert recognition, high-quality recipe generation with remarkably low perplexity, development and public release of comprehensive datasets, establishment of a scalable framework adaptable to other cultural preservation contexts, and demonstration of practical applications for endangered cultural knowledge preservation.

This work not only provides a viable method for the digital preservation of culinary heritage but also contributes a valuable, specialized dataset to the research community. The public availability of our datasets and methodologies ensures that other researchers can build upon this foundation, potentially extending the approach to other cultural domains and regional contexts.

Future work will focus on expanding the dataset with more recipe variations, exploring larger language models, and deploying the final system as a publicly accessible web application. Through such collaborative efforts, computational methods can play an increasingly important role in safeguarding cultural knowledge for future generations, ultimately contributing to the broader goal of maintaining humanity’s diverse cultural heritage in the digital age.

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